End-Semester Examination - 2024 (Autumn)
MATUGMCC1101 & MATUGMIN1101
(Differential and Integral Calculus)
Semester – I (4 Year B.Sc. Honours in Mathematics, Statistics & Physics) 3×3

Full Marks – 80 **Time – 3 Hours**

Notations and symbols have their usual meaning.

Section A. Answer ALL questions. Each question carries one mark.

1. The envelope of the family of straight lines $y = mx + \frac{5}{m}$ is
   - (a) $y^2 = 10x$
   - (b) $y^2 = 20x$
   - (c) $y^2 = 5x$
   - (d) $y^2 = x$

2. The point on the curve $\sqrt{x} + \sqrt{y} = 2$, where the tangent is equally inclined to the co-ordinate axis is
   - (a) $(1,2)$
   - (b) $(1,1)$
   - (c) $(0,4)$
   - (d) $(2,5)$

4×4

3. The pedal equation of $r = e^{\theta}$ is
   - (a) $p^2 = r$
   - (b) $p^2 = r^2$
   - (c) $2p^2 = r^2$
   - (d) $p^2 = 2r^2$

$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2[\sin\phi_1 \sin\phi_2 \cos(\theta_1 - \theta_2) + \cos\phi_1 \cos\phi_2}$

3×4

4. The number of normals to the curve $x^2 = 4y$ which pass through $(1,2)$ is
   - (a) 0
   - (b) 1
   - (c) 2
   - (d) more than 2

4×3

5. The radius of curvature of the curve $x + y = 1$ is
   - (a) 0
   - (b) 1
   - (c) 2
   - (d) ∞

5

6. If $f(2a-x)=f(x)$ then $\int_0^{2a} f(x)dx =$
   (a) $2\int_0^a f(x)dx$
   (b) $2\int_0^a f(-x)dx$
   (c) $\int_0^a f(x)dx$
   (d) 0

7. The area bounded by the curve $r=f(\theta)$ and the radii vectors $\theta=\alpha$ and $\theta=\beta$ is given by
   (a) $\int_\alpha^\beta r^2 d\theta$
   (b) $\frac{1}{2}\int_\alpha^\beta r^2 d\theta$
   (c) $\int_\alpha^\beta r d\theta$
   (d) $\frac{1}{2}\int_\alpha^\beta r d\theta$

8. If $I_n=\int_0^{\pi/2} \cos^n x dx$ and $I_n=\lambda I_{n-2}$ then $\lambda =$
   (a) $\frac{n}{n+1}$
   (b) $\frac{n}{n-1}$
   (c) $\frac{n-1}{n}$
   (d) $\frac{n+1}{n}$

9. The length of the arc of the curve $y=\cosh x$ between $x=0$ and $x=1$ is
   (a) $\cosh 1$
   (b) $\cos 1$
   (c) $\sinh 1$
   (d) $\sin 1$

10. The volume generated by revolving about $x-$axis, the area bounded by $y=x^3$ between $x=0$ and $x=1$ is
   (a) $\frac{\pi}{7}$
   (b) $\frac{2\pi}{7}$
   (c) $\frac{\pi}{6}$
   (d) $\frac{\pi}{5}$

Section B. Answer ALL questions. Each question carries one mark.

11. Find the $n-$th derivative of the function $y=(a-bx)^m$.

12. If $f(x)=x^n$, prove that $f(1)+\frac{f'(1)}{1!}+\frac{f''(1)}{2!}+\cdots+\frac{f^{(n)}(1)}{n!}=2^n$.

13. Show that the radius of curvature of a circle at any point is a constant.

14. Show that the chord of curvature parallel to the axis of y of the curve $y=a\log\sec(x/a)$ is a constant.

$\sin^2 x$ $\frac{\sin 2x}{2}$

15. Show that for the curve $r = e^{\theta}$, the polar subtangent is equal to the polar subnormal.
16. Trace the curve $r = a\cos 2\theta$.
17. Prove that $\int_a^b f(x)dx = \int_a^b f(a+b-x)$.
18. Prove that the area of the region bounded by the parabola $y^2 = 4x$ and its latus rectum is $\frac{8}{3}$.
19. What is the intrinsic equation of $y = c\cosh\frac{x}{c}$?
20. Find the surface area of the whole surface generated by revolving the circle $x^2 + y^2 = 1$ round the x-axis.

Section C. Answer any SIX questions. Each question carries 10 marks.

21. (a) Prove that $D^n(\tan^{-1}x) = (-1)^n(n-1)!\sin^n\theta\sin n\theta$, where $\cot\theta = x$. 5

    (b) State Leibnitz's theorem for the $n-$th derivative of the product of two functions. Use it to prove $y_n = \frac{(n-1)!}{x}$, where $y = x^{n-1}\log x$. 5

22. (a) If $y = \tan^{-1}x$, show that $(1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$. Also, find $y_n(0)$. 5

    (b) Obtain the reduction formula for $\int_0^{\frac{\pi}{2}}\sin^m x\cos^n x dx$. Use it to evaluate $\int_0^{\frac{\pi}{2}}\cos^4 x\sin^2 x dx$. 5

23. (a) If ρ_1 and ρ_2 be the radii of curvature at the ends P and D of conjugate diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then prove that $(ab)^{\frac{2}{3}}(\rho_1^{\frac{2}{3}} + \rho_2^{\frac{2}{3}}) = a^2 + b^2$. 5

    (b) Show that the pedal equation of the astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ is $r^2 + 3p^2 = a^2$. 5

24. (a) Prove that the envelope of the parabola $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$, where $ab = k^2$ is $16xy = k^2$, where a and b are variable parameters. 5

    (b) Determine the asymptotes of the cubic $x^3 - 2y^3 + xy(2x-y) + y(x-y) + 1 = 0$. 5

25. (a) Show that the curves $\frac{x^2}{a} + \frac{y^2}{b} = 1$ and $\frac{x^2}{a'} + \frac{y^2}{b'} = 1$ will cut orthogonally if $a - b = a' - b'$. 5

    (b) If ϕ be the angle between the tangent and radius vector at any point on the curve $r = f(\theta)$, then prove that $\tan\phi = r\frac{d\theta}{dr}$. 5

26. (a) Find the area of one loop of the curve $r = a\cos 2\theta$. 5

    (b) Find the area of the portion of the circle $x^2 + y^2 = 1$, which lies inside the parabola $y^2 = 1 - x$. 5

27. (a) If s be the length of the curve $r = a\tanh\frac{1}{2}\theta$ between the origin and $\theta = 2\pi$ and Δ the area between the same points, show that $\Delta = a(s - a\pi)$. 5

    (b) Show that in the cycloid $x = a(\theta + \sin\theta)$, $y = a(1-\cos\theta)$, $\rho^2 + s^2 = 16a^2$, the arc being measured from the vertex. 5

28. (a) Show that the volume of the solid produced by the revolution of the loop of the curve $y^2(a+x) = (a-x)$ about the x-axis is $2\pi a^3(\log 2 - \frac{2}{3})$. 5

$(1+x^2)^{-1}$ x^{n-1}

(b) The portion between the two consecutive cusps of the cycloid
$x = a(\theta + \sin\theta)$, $y = a(1 + \cos\theta)$ is revolved about the x-axis. Show that the ratio of the area of the surface so formed and the area of the cycloid is $64:9$. 5

29. (a) Evaluate $\lim_{n\to\infty}\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\cdots\left(1+\frac{n}{n}\right)\right]^{\frac{1}{n}}$ 5

(b) Prove that $\int_0^\pi \frac{x\,dx}{a^2\sin^2 x + b^2\cos^2 x} = \frac{\pi^2}{2ab}, (a, b > 0)$. 5

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